

Wind, fetch and current: the physics of wave steepness

Working notes for the Unified Marine Map (steepness layer) · draft, revised as we go

This note explains, for a curious non-physicist, how the map turns a wind forecast and a current forecast into a **steepness** hazard — and why the answer is often counter-intuitive: a modest shift in wind direction, or a small change in current, can matter more than a big change in wind speed. Every equation below is the one the live map actually evaluates, and the two charts are drawn from those same formulas. It closes with the open question we want to explore next: the **force** a steep wave puts on a kayak.

1 • It is steepness, not height, that tips a boat

A wave's danger is better measured by its **steepness** — height divided by wavelength — than by its height alone.

$S = H/L$ a 1 m swell 100 m long ($S = 1/100$) is a gentle roll; a 1 m wave 7 m long ($S = 1/7$) is a breaking wall

Deep-water gravity waves become unstable and break when steepness reaches the limiting value found by Michell (1893) and Stokes for the highest progressive wave, where the crest sharpens to a 120° angle:

$S_{\max} = H/L \approx 1/7 = 0.143$ in a real irregular sea a significant steepness near 1/10 already means many crests are breaking

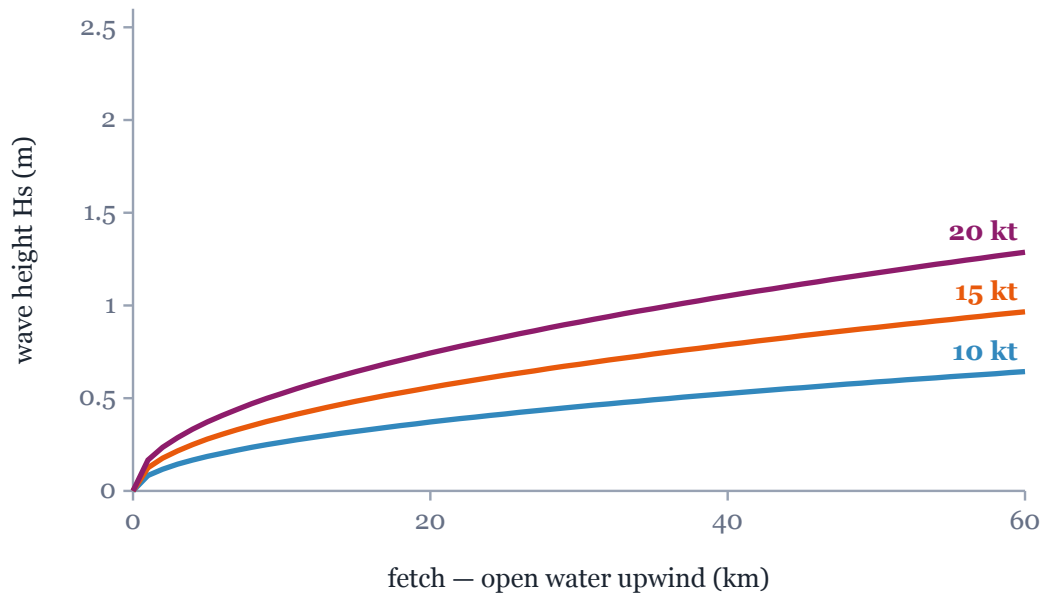
The map therefore reports steepness in four bands: **mild** ($S \approx 0.05$), **caution** (≈ 0.075), **rough** (≈ 0.10) and **hazardous / breaking** (≈ 0.143).

2 • How wind builds a sea — and why fetch is the lever

Wind pushes on the water and waves grow. Two things cap how big they get: the **wind speed**, and the **fetch** — the distance of open water the wind crosses before reaching you. Near an upwind shore the sea is small; the further open water stretches upwind, the more the waves build, until they reach a “fully developed” size for that wind and grow no more. A breeze down the length of a strait raises a real sea; the same breeze across a narrow channel raises only ripples.

That is why the map computes fetch **directionally**: at every water cell it looks upwind and measures how far open water runs before it hits land, separately for **16 compass directions**.

When the forecast wind blows from a given direction, the map uses the fetch in *that* direction — so the same wind speed produces very different seas depending only on which way it blows.



Wave height vs fetch, from the map's growth formula. Each curve flattens at its fully-developed ceiling.

3 • How the wave height (H_s) and period (T_p) are derived

The map uses the standard **fetch-limited growth laws** — the JONSWAP field-experiment fit (Hasselmann et al., 1973) — capped at the **Pierson–Moskowitz** fully-developed sea (Pierson & Moskowitz, 1964). With wind speed U (m/s), fetch F (m) and gravity $g = 9.81 \text{ m/s}^2$:

$$H_s = 4 \cdot U \cdot \sqrt{(1.6 \times 10^{-7} \cdot F / g)} \quad \text{significant wave height — grows with wind, and with the square-root of fetch}$$

$$T_p = 1 / [3.5 \cdot (gU) \cdot (gFU^2)^{-0.33}] \quad \text{peak period — longer fetch gives longer, faster waves}$$

$$H_s \leq 0.24 \cdot U^2 / g, \quad T_p \leq 0.72 \cdot U \quad \text{fully-developed ceilings (Pierson–Moskowitz) — the flat tops in the chart above}$$

The subtle point: **height and period both rise with fetch, but the period rises faster**. A longer fetch makes waves taller *and* longer — and longer waves are less steep and harder for a current to trip. So a bigger sea is not automatically a rougher one, as §5 shows.

4 • How an opposing current steepens — and blocks — the sea

A deep-water wave travels at a speed set by its period, and carries a length to match:

$$c_0 = g \cdot T_p / 2\pi \text{ (wave speed) , } L_0 = c_0 \cdot T_p \text{ (wavelength) } \text{ initial steepness } S_0 = H_s / L_0$$

Only the part of the current pointing *into* the waves matters. With θ the angle between the current's set and the direction the wind blows from,

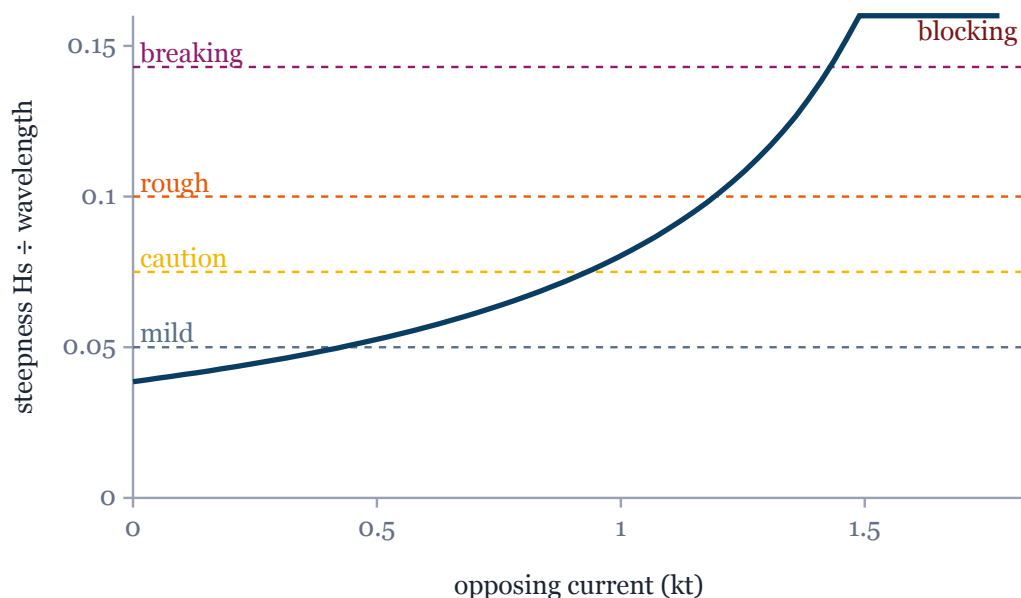
$$U_{\text{opp}} = V_{\text{current}} \cdot \cos \theta \text{ head-on is fully opposing; a cross-current barely counts; a following current does nothing}$$

A wave riding onto an opposing current keeps its period but is forced to slow and **shorten**; conserving its wave-action it grows **taller** (Longuet-Higgins & Stewart, 1961; Phillips, 1977). Both effects raise steepness. Writing $\alpha = -U_{\text{opp}}/c_0$ and the shortening factor $x = k_0/k$ from the wave-on-current dispersion relation, the steepness climbs as

$$S \approx (x^{3.5} / \sqrt{2-x}) \cdot S_0$$

When the opposing current reaches **a quarter of the wave speed** the waves can no longer advance; they stop, pile up and break. This is the classic wave-blocking limit (Chawla & Kirby, 2002):

$$U_{\text{opp}} \geq c_0 / 4 \text{ short (slow) waves have a low } c_0, \text{ so they block at a modest current — the reason short-fetch chop is dangerous}$$



Steepness of the example sea (0.35 m @ 2.4 s) as an opposing current builds — it runs away toward breaking as the current nears the blocking speed.

5 • Worked example — small shifts, big change

Same 15-knot wind at one spot beside a channel. The wind **veers about 25°** so that instead of blowing along the channel it blows across it — the upwind fetch drops from 16 km to 8 km — and at the same time the ebb **strengthens from 1.0 to 1.2 knots** against the waves. Neither change is large:

	A • along-channel	B • across-channel
Wind	15 kt	15 kt
Fetch upwind	16 km	8 km
Wave height H_s	0.50 m	0.35 m
Wave period T_p	3.0 s	2.4 s
Opposing current	1.0 kt	1.2 kt
Blocking current (co4)	2.3 kt	1.8 kt
Current ÷ blocking	44 %	66 %
Steepness S	0.060	0.102
Result	mild	rough

The across-channel waves are actually **smaller** (0.35 m vs 0.50 m) — yet the map flags them far worse. Being shorter and slower, they block at a lower current, and the modest extra ebb pushes them from 44% to 66% of that limit. The hazard is not the wave size; it is **how steep the wind and current together make it**.

6 • The open question: from steepness to *force*

Steepness tells you a wave is likely to break and stack up — but a paddler feels a **force**. Two physically grounded quantities scale that force, and both rise fast with height:

Energy density $E = \frac{1}{8} \cdot \rho \cdot g \cdot H^2$ energy per unit sea-surface area — scales with the square of height

Energy flux (power) $P = E \cdot c_g \propto H^2 \cdot T$ rate energy is delivered — rises with height squared AND period
(group speed $c_g = gT/4\pi$)

So a wave twice as high carries roughly **four times** the energy, and a longer period delivers it faster — which is the grain of truth in the instructor’s “force scales sharply with height and period.” The peak **impact** of a breaking or near-breaking wave on a hull is harder: it depends on the water-particle velocity at the crest (itself $\propto$ steepness $\times$ celerity) and on slamming dynamics studied in coastal engineering (Goda, 2000; Cuomo et al., 2010). Building a defensible, literature-validated **force/energy index** — and an interactive graph of how wind direction, fetch and current move both steepness and that index together — is the next step. This document will grow to cover it.

Status. The steepness layer is experimental and **not yet ground-truthed**. These notes are for understanding and refining the model before it is shown publicly. The force index in §6 is not yet in the map.

References

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